Daniel J Noh — (he/him)

Ph.D. Student, University of Pennsylvania

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Research Overview

learning sciences, care, design + computing education, participatory design, human-computer interaction, algorithmic justice, constructionism, design of learning spaces, maintenance + repair, accessibility

Education

Expected 2029	**University of Pennsylvania** *Ph.D.,* Learning Sciences & Technologies Advisor: Yasmin B. Kafai
2022	**Harvard University** *Ed.M.,* Learning Design, Innovation & Technology
2021	**Carnegie Mellon University** *B.Arch*., Architectural & Building Sciences/Technology (University & College Honors) Minors: Human-Computer Interaction & Design for Learning

Research Experience

Aug 2024 to current — **University of Pennsylvania**
Graduate School of Education — Kafai Lab (PI: Yasmin Kafai)

NSF Grants #2333469, #2342438, #2414590
Spencer, Kapor, William T. Grant, Alfred P. Sloan Foundations Rapid Response Funding
Google Academic Research Award: AI for Privacy, Safety, & Security (2025)

Research Fellow

- Collecting and analyzing data to investigate learners' and educators' conceptions of AI/ML systems, including generative language models, social media, and advertisements, and algorithmic justice in formal and informal learning environments
- Developing related learning tools and activities through participatory design with educators and youth

Jun 2024 to Aug 2024 — **Northeastern University**
College of Arts, Media and Design (PI: Sara Hendren)

Research Assistant

- Developed experimental pedagogical modules for the studio-seminar course, *Investigating Normal: Design and Disability*, identifying best practices for accessible spatial and digital design, related precedent projects, and assessment strategies for proposed activities and curricular prompts

Sep 2021 to Jun 2022 — **Harvard University**
Project Zero — Designing Learning Places Lab (PI: Daniel Wilson)

Research Assistant

- Conducted a literature review on the qualities of materials, objects, and environments that support the learning practices of noticing, wondering, and help-seeking
- Designed a practical toolkit for practitioners to adopt the findings into the design of learning spaces
- Co-developed the course, *Designing Learning Places*, currently taught at the Harvard Graduate School of Education

Sep 2020 to Sep 2021 | **Carnegie Mellon University**
Human-Computer Interaction Institute — ClassInSight (PI: Amy Ogan)

James S. McDonnell Foundation Grant #1822813

Research Associate (May 2021 to Sep 2021)
Research Assistant (Sep 2020 to May 2021)

- Drafted and conducted focus group protocols, interview questions, and participatory design sessions to understand how classroom discussion data visualizations and personalized professional development can scaffold teacher agency and meaning-making
- Designed prototypes and data visualizations for a conversation support professional development tool

Honors & Awards

2024-2028 | **Fontaine Society Fellowship,** University of Pennsylvania
2021 | **GEE! Learning Games Award Finalist,** GEE! Learning Games Awards
2020 | **Askwith Kenner Artist Grant,** CMU Department of Modern Languages
2020 | **Louis F. Valentour Scholarship,** CMU School of Architecture

Publications

Journal Articles (Peer-Reviewed)

[J.1] Morales-Navarro, L., **Noh, D. J.**, & Kafai, Y. B. (2025). High school students building babyGPTs: Engaging in data practices and addressing ethical issues through the construction of generative language models. *International Journal of Child-Computer Interaction, 45*, 100769.
DOI: https://doi.org/10.1016/j.ijcci.2025.100769

Conference Proceedings (Peer-Reviewed, Full)

[C.4] Morales-Navarro, L., **Noh, D. J.**, Servat, L., Netting, C., Kafai, Y. B., & Metaxa, D. (2026, *Accepted*). Building to Understand: Examining Teens' Technical and Socio-Ethical Pieces of Understanding in the Construction of Small Generative Language Models. In *Proceedings of the 25th Interaction Design and Children.*
DOI: https://doi.org/10.1145/3773077.3806107

[C.3] **Noh, D. J.**, Fields, D. A., Kafai, Y. B., & Metaxa, D. (2026, *Accepted*). "You Can Actually Do Something": Shifts in High School Computer Science Teachers' Conceptions of AI/ML Systems and Algorithmic Justice. In *Proceedings of the 20th International Conference of the Learning Sciences.* International Society of the Learning Sciences.

[C.2] **Noh, D. J.** (2026). Guide on the Side or Sage on the Stage?: Exploring the Relationship Between Teachers' Spatial and Verbal Discursive Strategies. In G. Carmona, C. Lima, M. J. Santos, H. Benítez, L. Montero-Moguel, & B. Galarza-Tohen (Eds.), *Advances in Quantitative Ethnography* (Vol. 2677, pp. 509–523). Springer Nature Switzerland.
DOI: https://doi.org/10.1007/978-3-032-12229-2_33

[C.1] **Noh, D. J.**, Fields, D. A., Morales-Navarro, L., Cabrera-Sutch, A. M., Kafai, Y. B., & Metaxa, D. (2025). Youth as Advisors in Participatory Design: Situating Teens' Expertise in Everyday Algorithm Auditing with Teachers and Researchers. In *Proceedings of the 24th Interaction Design and Children*, 415–428.
DOI: https://doi.org/10.1145/3713043.3728849 [Acceptance Rate: 29%]

Conference Proceedings (Peer-Reviewed, Short & Symposia)

[S.5] Morales-Navarro, L., Fields, D., Giang, M. T., **Noh, D. J.**, Kafai, Y. & Metaxa, D. (2026, Accepted). Understanding teens' self-beliefs when learning to construct and deconstruct AI/ML systems: Developing a survey instrument. In *Proceedings of the 25th Interaction Design and Children.* DOI: https://doi.org/10.1145/3773077.3812156

[S.4] Morales-Navarro, L., Kafai, Y., Hilke, O., Pope, N., Tedre, M., **Noh, D. J.**, Fields, D. A., Metaxa, D., Solyst, J., Shapiro, R. B., Fong, C., Nurdin, F., Sharma, S., Iivari, N., Klemettilä, P. A., Bilstrup, K-E., Musaeus, L. H., Landesman, R., Everson, J., Ryoo, J. & Iversen, O. S. (2026, Accepted). Learning to Deconstruct AI/ML Systems: Scaffolds to Support Learners in Critically Evaluating AI/ML Systems. In *Proceedings of the 20th International Conference of the Learning Sciences.* International Society of the Learning Sciences.

[S.3] Ching, C., Cortez, A., Higgs, J., Grover, S., Vakil, S., Paik, W., Abdul, S., Knight, M., Ryoo, J., Choi, M., Wei, W., Clarke, R., Blizzard-Caron, J., Stark, L., Kohn, L., Voloch, D., Ochterski, J., Lizárraga, J., Wang, X. C., Xing, G., Dantu, K., Morales-Navarro, L., **Noh, D.**, Kafai, Y. & Stornaiuolu, A. (2026, Accepted). Artificial Intelligence and Human Purposes: Dialogues about Creativity and Control. In *Proceedings of the 20th International Conference of the Learning Sciences.* International Society of the Learning Sciences.

[S.2] Morales-Navarro, L., **Noh, D. J.**, & Kafai, Y. (2025). Building babyGPTs: Youth engaging in data practices and ethical considerations through the construction of generative language models. In *Proceedings of the 24th Interaction Design and Children*, 1021–1026. DOI: https://doi.org/10.1145/3713043.3731525

[S.1] Kafai, Y., Shapiro, R. B., Jetzinger, F., Michaeli, T., Tedre, M., Vartiainen, H., Iivari, N., Musaeus, L. H., Iversen, O. S., Ali, S., Bodon, H., Butler, M., Kshirsagar, K., Smith, M., Quiterio, A., Worsley, M., Kumar, V., Morales-Navarro, L., **Noh, D.**, Pea, R., & Philip, T. M. (2025). Youth as Designers of Artificial Intelligence and Machine Learning Technologies: What Do We Know About the Opportunities and Challenges of K-12 Students Creating Their Own Applications?. In *Proceedings of the 19th International Conference of the Learning Sciences*, 2260-2268. International Society of the Learning Sciences. DOI: https://doi.org/10.22318/icls2025.362276

Posters, Presentations & Workshops (Peer-Reviewed)

[P.7] Porter, B., Okker-Edging, K., Liu, A., **Noh, D. J.**, Lydic, K., Muzekari, B., Lewis, T., Torres-Grillo, O., Falk, E., Tan, A. S.L. & Cosme, D. (2026, May). Youth Participatory Action Communication Research on Social and Environmental Determinants of Health. *2026 Conference for Advancing Participatory Sciences.* Virtual.

[P.6] **Noh, D.** Fields, D. A., Kafai, Y. B. & Metaxa, D. (2026, April). "A Path to Activism": CS Teachers Developing Critical Agency with Algorithmic Systems through Algorithm Auditing. *American Educational Research Association (AERA) Conference 2026.* Los Angeles, CA.

[P.5] Morales-Navarro, L., **Noh, D.** & Kafai, Y. B. (2026, April). High School Students as Designers of Generative Language Models. *American Educational Research Association (AERA) Conference 2026.* Los Angeles, CA.

[P.4] Morales-Navarro, L., **Noh, D.** & Kafai, Y. B. (2026, April). Teens' Ethical Considerations When Building babyGPTs. *American Educational Research Association (AERA) Conference 2026.* Los Angeles, CA.

[P.3] Fields, D. A., Morales-Navarro, L., **Noh, D.**, Kafai, Y. B. & Ottina, J. (2025, October). Workshop: Investigating the Black Box of Artificial Intelligence and Machine Learning: Promoting Youth and Teacher Agency through Algorithm Auditing. *Connected Learning Summit 2025.* Virtual.

[P.2] **Noh, D. J.**, Morales-Navarro, L., & Kafai, Y. (2025, October). What Comes Next?: Youth Learning About and Designing Markov Chains through Unplugged Activities. *Connected Learning Summit 2025.* Virtual.

[P.1] Morales-Navarro, L., **Noh, D. J.**, & Kafai, Y. (2025, October). What happens when teens design small generative language models? A case study of teenagers building babyGPTs. *Connected Learning Summit 2025.* Virtual.

Invited Talks & Presentations

[T.1] **Noh, D. J.** (2026, April). Youth as Advisors and Teachers as Designers: The Participatory Design of AI Auditing Lessons. Invited presentation and workshop for *Penn GSE AI & Education Symposium.* Philadelphia, PA.

Public Scholarship (White Papers, Curriculum & Other)

[PS.2] Butapetch, H., Fields, D., Hanna, S., Landa. J., Morales-Navarro, L., Myers, T., **Noh, D.**, Ottina, J., & Ulrich, A., with Kafai, Y. & Metaxa. D. (2025). Introduction to Algorithm Auditing: Lessons for High School Computer Science (beta version).
Available at: https://bit.ly/k12aiaudit-lessons

[PS.1] Gonzalez, P. G., **Noh, D.**, & Wilson, D. (2022). Making the Space for Learning. *Presidents and Fellows of Harvard College. Cambridge, MA.*
Available at: https://pz.harvard.edu/resources/making-the-space-for-learning

News and Press

[N.2] Building babyGPTs: How Penn GSE and the Franklin Institute are redefining AI literacy for youth. (2025, August). *PennGSE News.*
Available at: https://penntoday.upenn.edu/news/penn-gse-building-babygpts

[N.1] Bringing Design Education to K-12 Students. (2023, April). *MIT MAD News.*
Available at: https://design.mit.edu/news/bringing-design-education-to-k-12-students

Teaching

Jan 2022 to Jan 2025

Harvard University
Graduate School of Education
**As a Teaching Fellow I was involved in course redesign, development of teaching material, leading weekly in-class lectures/workshops, and providing constructive student feedback*

Teaching Fellow, **Rapid Prototyping of Educational Products**
Instructor: Bertrand Schneider (Winter 2025)

Lead Teaching Fellow & Course Co-Designer, **Designing Learning Places**
Instructor: Daniel Wilson (Spring 2023, Spring 2024)

Teaching Fellow, **Digital Fabrication and Making in Education**
Instructor: Bertrand Schneider (Spring 2022, Fall 2022, Spring 2024)

Teaching Fellow, **Transforming Education Through Emerging Technologies**
Instructor: Bertrand Schneider (Spring 2024)

Teaching Fellow, **Multimodal Learning Analytics**
Instructor: Bertrand Schneider (Spring 2023)

Feb 2024 to May 2024

Design Museum Everywhere
High School Inclusive Design Challenge (Co-Instructor: Rosa Weinberg)

Design Educator
- Planned and facilitated a three-week long program on accessibility, inclusive design, and participatory design with six financially-compensated high school students

Sep 2022 to Jan 2024 | **Massachusetts Institute of Technology**
MIT Museum

Technical Instructor / Museum Educator
- Designed and facilitated novel hands-on STEAM learning experiences and public events to build meaningful connections to scientific inquiry and design exploration
- Established standard operating procedures and trained museum educators on various fabrication tools in the makerspace (laser cutter, 3D printers, and vinyl cutters)
- Mentored MIT undergraduate research assistants (UROP) and museum teen interns on the development and facilitation of design-based, hands-on learning experiences
- Co-led the launch and development of the Design Redefined series, co-hosted by MIT MAD and Innovators for Purpose

Advising & Mentoring
* = Indicates co-authorship in publication

Undergraduate Researchers

- 2026 — Sophia Wang (University of Pennsylvania)
- 2025-2026 — Lucianne Servat* (University of Pennsylvania, with Luis Morales-Navarro)
- 2025 — Elo Esalomi (University of Pennsylvania, with Luis Morales-Navarro)
- 2023-2024 — Wonuola Abiodun (MIT, with Rosa Weinberg)
- 2023-2024 — Alexandra Coston (MIT, with Rosa Weinberg)
- 2023 — Kimberly McPherson (MIT, with Rosa Weinberg)

Service

Journal Reviewing | Journal of the Learning Sciences (2026)

Conference Reviewing | ACM Interaction Design & Children (2026)
International Conference of the Learning Sciences (2026)
Educational Advances in Artificial Intelligence (2026)
Connected Learning Summit (2025)
International Conference on Quantitative Ethnography (2025)

Institutional Service | **University of Pennsylvania**
- Penn GSE, Board of Advisors Student Presenter (2026)

Massachusetts Institute of Technology
- MIT-Nord Anglia, STEAM Teacher Professional Development Mentor (2024)

Carnegie Mellon University
- School of Architecture, Student Portfolio Reviewer (2021)
- School of Architecture, Architecture Peer Mentor (2017)

Other Service | **Carnegie Museum of Natural History**
- Museum Volunteer (2020)

Other Experience

Aug 2021 to May 2022 | **Harvard University**
Graduate School of Education – Education Innovation Studio

Makerspace Assistant
- Taught graduate students how to use fabrication tools including laser cutters, 3D printers, vinyl cutters, sewing machines, and power tools

May 2020 to Aug 2020 | **Carnegie Museum of Natural History**
Center for Anthropocene Studies (Advisors: Asia Ward & Marti Louw)

Center for Anthropocene Studies Education Intern
- Produced educational videos about sewage treatment and potable water, presented at ALCOSAN's 2020 and 2021 Open House
- Published corresponding articles in the museum's monthly newsletters and website

May 2018 to Jan 2021 | **Carnegie Mellon University**
School of Architecture

Design Fabrication (dFab) Lab Assistant
- Assisted architecture students and faculty on implementing digital fabrication methods (laser cutters, 3D printers, CNC-mill, and vacuum formers) into their project workflow

Feb 2019 to May 2020 | **Carnegie Mellon University**
School of Architecture, Manufacturing Futures Initiative (PI: Joshua Bard)

Research Assistant
- Investigated the use of steam-bending and novel joinery to design, prototype, and fabricate steam-bent, wooden swings for a public park in Pittsburgh

Skills & Tools

Programming	Python, JavaScript, HTML/CSS, R
Fabrication	Laser Cutting, 3D Printing, Vinyl Cutting, Vacuum Forming, Sewing, Woodworking, Soldering, Basic Electronics/Microcontrollers
Design	Rapid Prototyping, Wireframing, Storyboarding, Illustrating, Data Visualization, Graphic Design, Architectural Design, Learning Design, Video Editing, Web Design & Development
Methods	Thematic Analysis, Interviews, Participatory Design Research, Youth Participatory Action Research, Multi-Modal Learning Analytics, Epistemic/Ordered Network Analysis

Professional Memberships

American Education Research Association (AERA)
International Society of the Learning Sciences (ISLS) – DIV C
Association of Computer Machinery (ACM) – SIGCHI